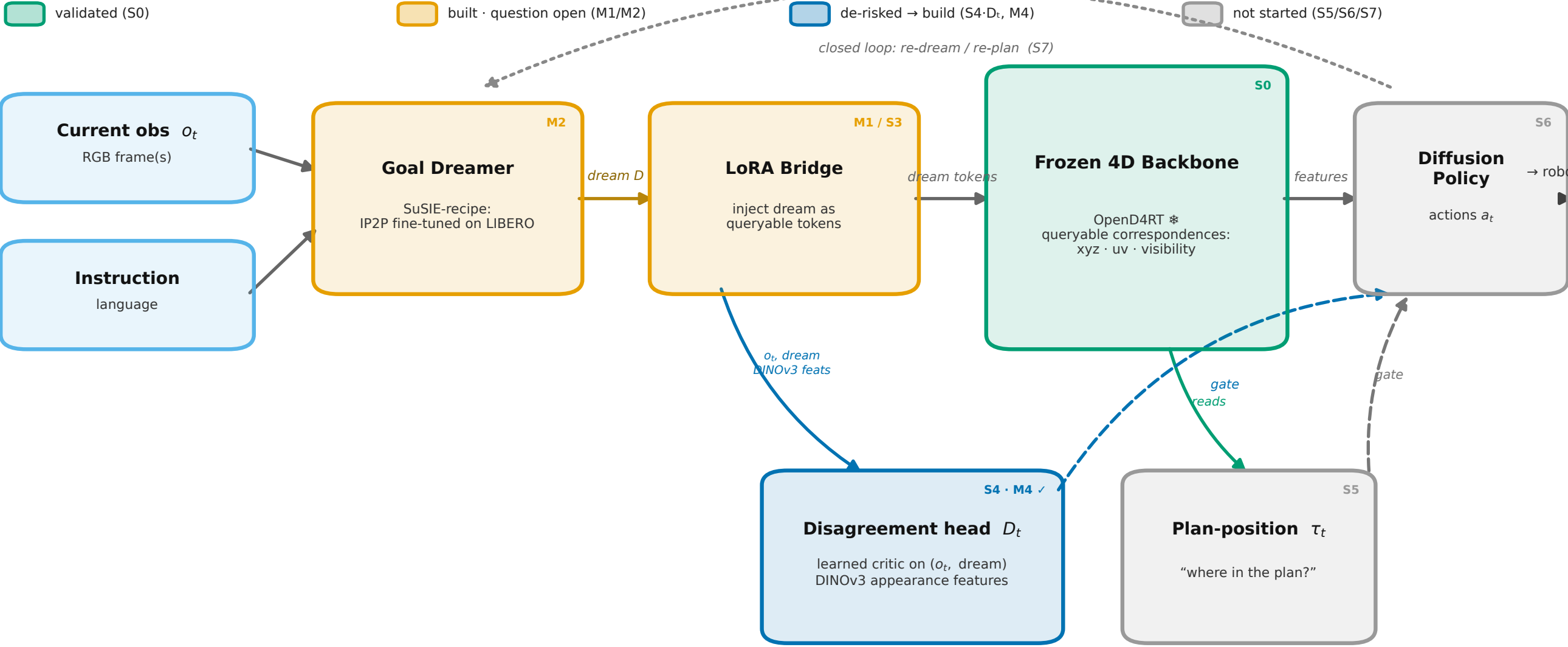


GD-4D — Grounded Dreaming on a Frozen 4D Backbone

goal-image-conditioned robot policy; the dreamed goal is injected into a queryable 4D reconstruction, and two gates (D_t , τ_t) modulate the policy



✓ D_t resolved — the geometric read-out is falsified; D_t is a **LEARNED CRITIC** (de-risked viable, M4)

M1/M2/M3 falsified the frozen-backbone GEOMETRIC signal (self/cross-consistency at chance on real dreams: M2 delta 0.51, M3 ~0.60 on a plausible-but-wrong future). M4: a learned critic on DINOv3 appearance features of (o_t , dream) separates correct vs wrong goals at AUROC 0.83–0.98 and transfers to generated dreams (0.78). $\Rightarrow D_t$ = the plan's S4-supervised appearance critic, NOT a backbone read-out; the frozen backbone's role narrows to dream-conditioning. Open: within-scene wrongness (in progress).